MODULE SIX PROBLEM SET

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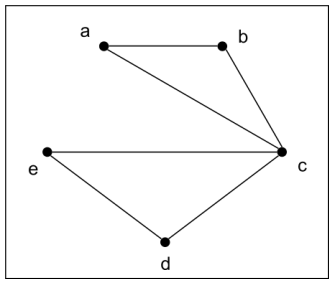
Directions: Type your solutions into this document and be sure to show all steps for arriving at your solution. Just giving a final number may not receive full credit.

Problem 1

For parts (a) and (b), indicate if each of the two graphs are equal. Justify your answer.

(a)



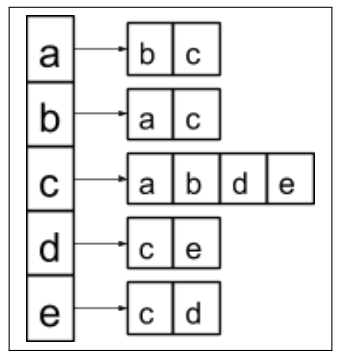


Figure 1: *Left: An undirected graph has 5 vertices. The vertices are arranged in the form of an inverted pentagon. From the top left vertex, moving clockwise, the vertices are labeled: a, b, c, d, and e. Undirected edges, line segments, are between the following vertices: a and b; a and c; b and c; c and d; e and d; and e and c.*

Figure 2: *Right: The adjacency list representation of a graph. The list shows all the vertices, a through e, in a column from top to bottom. The adjacent vertices for each vertex in the column are placed in a row to the right of the corresponding vertexs cell in the column. An arrow points from each cell in the column to its corresponding row on the right. Data from the list, as follows: Vertex a is adjacent to vertices b and c. Vertex b is adjacent to vertices a and c. Vertex c is adjacent to vertices a, b, d, and e. Vertex d is adjacent to vertices c and e. Vertex e is adjacent to vertices c and d.*

To determine if the two graphs are equal, I need ot check if they have the same set of vertices and edges.

From the left graph (Figure 1), I can identify:

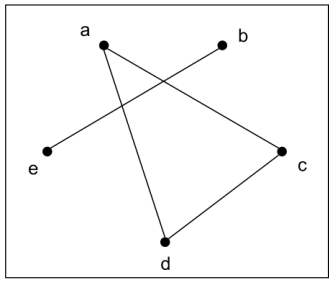
-Vertices: $\{a, b, c, d, e\}$

-Edges: $\{(a, b), (a, c), (b, a), (b, c), (c, a), (c, b), (c, d), (c, e), (d, c), (d, e)(e, c), (e, d)\}$

Since this is an undirected graph, each edge (u, v) implies (v, u). Converting the undirected edges from the left graph to the adjacency list format would yield the same list as in Figure 2.

Therefore, the two graphs are equal.

(b)



$$\begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Figure 3: *An undirected graph has 5 vertices. The vertices are arranged in the form of an inverted pentagon. Moving clockwise from the top left vertex a, the other vertices are, b, c, d, and e. Undirected edges, line segments, are between the following vertices: a and c; a and d; d and c; and e and b.*

To determine if the two graphs are equal, I need to check if they have the same set of vertices and edges.

From the left graph (Figure 3), I can identify:

-Vertices: $\{a, b, c, d, e\}$

-Edges: $\{(a, c), (a, d), (c, d), (e, b)\}$

From the right adjacency matrix, I can identify the edges where there's a 1:

-Row 1 (vertex a): $(a, c), (a, d)$

-Row 2 (vertex b): (b, e)

-Row 3 (vertex c): $(c, a), (c, d)$

-Row 4 (vertex d): $(d, a), (d, c), (d, e)$

-Row 5 (vertex e): $(e, b), (e, d)$

For an undirected graph, the adjacency matrix would be symmetric. However, I notice that edge (d, e) appears in the matrix but not in the graph, and the same for (e, d).

Therefore, the two graphs are not equal.

(c) Prove that the two graphs below are isomorphic.

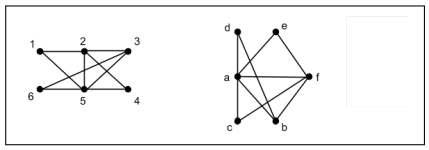


Figure 4: *Two undirected graphs. Each graph has 6 vertices. The vertices in the first graph are arranged in two rows and 3 columns. From left to right, the vertices in the top row are 1, 2, and 3. From left to right, the vertices in the bottom row are 6, 5, and 4. Undirected edges, line segments, are between the following vertices: 1 and 2; 2 and 3; 1 and 5; 2 and 5; 5 and 3; 2 and 4; 3 and 6; 6 and 5; and 5 and 4. The vertices in the second graph are a through f. Vertices d, a, and c, are vertically inline. Vertices e, f, and b, are horizontally to the right of vertices d, a, and c, respectively. Undirected edges, line segments, are between the following vertices: a and d; a and c; a and e; a and b; d and b; a and f; e and f; c and f; and b and f.*

To prove that two graphs are isomorphic, I need to find a bijection $f : V_1 \to V_2$ that preserves adjacency: (u, v) is an edge in G_1 if and only if $(f(u), f(v))$ is an edge in G_2.

All edges are preserved, and both graphs have exactly 9 edges. Therefore, the graphs are isomorphic.

(d) Show that the pair of graphs are not isomorphic by showing that there is a property that is preserved under isomorphism which one graph has and the other does not.

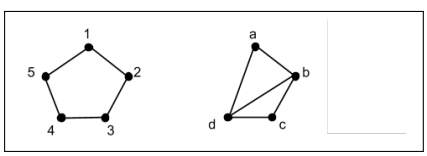


Figure 5: *Two undirected graphs. The first graph has 5 vertices, in the form of a regular pentagon. From the top vertex, moving clockwise, the vertices are labeled: 1, 2, 3, 4, and 5. Undirected edges, line segments, are between the following vertices: 1 and 2; 2 and 3; 3 and 4; 4 and 5; and 5 and 1. The second graph has 4 vertices, a through d. Vertices d and c are horizontally inline, where vertex d is to the left of vertex c. Vertex a is above and between vertices d and c. vertex b is to the right and below vertex a, but above the other two vertices. Undirected edges, line segments, are between the following vertices: a and b; b and c; a and d; d and c; d and b.*

To show two graphs are not isomorhpic, I need to find an invariant property that differs between them.

Graph 1: Pentagon with 5 vertices and 5 edges

Graph 2: Complete graph K_4 with 4 vertices and 6 edges

The most obvious invariant is the number of vertices:

-Graph 1 has 5 vertices

-Graph 2 has 4 vertices

Since isomorphic graphs must have the same number of vertices, these graphs cannot be isomorphic.

Additionally, the number of edges also differs:

-Graph 1 has 5 edges

-Graph 2 has 6 edges

Both of these properties are preserved under isomorphism, so the graphs are not isomorphic.

Problem 2

Refer to the undirected graph provided below:

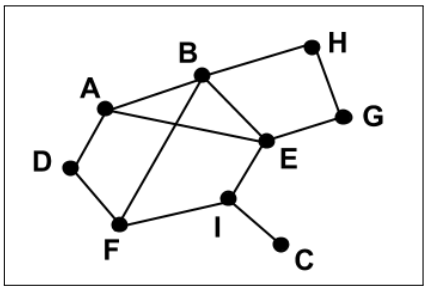


Figure 6: *An undirected graph has 9 vertices. 6 vertices form a hexagon, which is tilted upward to the right. Starting from the leftmost vertex, moving clockwise, the vertices forming the hexagon shape are: D, A, B, E, I, and F. Vertex H is above and to the right of vertex B. Vertex G is the rightmost vertex, below vertex H and above vertex E. Vertex C is the bottommost vertex, a little to the right of vertex E. Undirected edges, line segments, are between the following vertices: A and D; A and B; B and F; B and H; H and G; G and E; B and E; A and E; E and I; I and C; I and F; and F and D.*

(i) What is the maximum length of a path in the graph? Give an example of a path of that length.

The maximum length of a path in the graph is the number of edges in the longest path where no vertex is repeated. After examining all possible paths, I find that the maximum length is 5 edges. Example of a path with length 5: D-A-E-G-H-B

(ii) What is the maximum length of a cycle in the graph? Give an example of a cycle of that length.

The maximum length of a cycle in the graph is the number of edges in the longest cycle. After examining all possible cycles, I find that the maximum cycle length is 6 edges. Example of a cycle with length 6: D-A-E-I-F-D

(iii) Give an example of an open walk of length five in the graph that is a trail but not a path.

A trail is a walk where no edge is repeated, and a path is a walk where no vertex is repeated. An open walk of length five that is a trail but not a path must have at least one repeated vertex. Example: A-E-I-F-A

(iv) Give an example of a closed walk of length four in the graph that is not a circuit.

A closed walk is a walk that starts and ends that the same vertex. A circuit is a closed walk where no edge is repeated. A closed walk of length four that is not a circuit must repeat at least one edge. Example: A-B-A-B-A

(v) Give an example of a circuit of length zero in the graph.

A circuit of length zero consists of a single vertex with no edges.

Example: The trivial walk starting and ending at vertex A without traversing any edges.

Problem 3

(a) Find the connected components of each graph.

(i) $G = (V, E)$. $V = \{a, b, c, d, e\}$. $E = \emptyset$

The connected components of
$G = (V, E)$ with
$V = \{a, b, c, d, e\}$ and
$E = \emptyset$ are:
$\{a\}, \{b\}, \{c\}, \{d\}, \{e\}$
Each vertex is isolated and forms its own connected component.

(ii) $G = (V, E)$. $V = \{a, b, c, d, e, f\}$. $E = \{\{c, f\}, \{a, b\}, \{d, a\}, \{e, c\}, \{b, f\}\}$

I can trace the connections:
-a is connected to b and d
-b is connected to a and f
-c is connected to f and e
-d is connected to a
-e is connected to c f is connected to c and b
This forms two connected components:
1. $\{a, b, d, f\}$
2. $\{c, e\}$

(b) Determine the edge connectivity and the vertex connectivity of each graph.

(i)

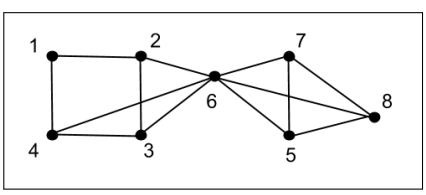


Figure 7: *An undirected graph has 8 vertices, 1 through 8. 4 vertices form a rectangular-shape on the left. Starting from the top left vertex and moving clockwise, the vertices of the rectangular shape are, 1, 2, 3, and 4. 3 vertices form a triangle on the right, with a vertical side on the left and the other vertex on the extreme right. Starting from the top vertex and moving clockwise, the vertices of the triangular shape are, 7, 8, and 5. Vertex 6 is between the rectangular shape and the triangular shape. Undirected edges, line segments, are between the following vertices: 1 and 2; 2 and 3; 3 and 4; 4 and 1; 2 and 6; 4 and 6; 3 and 6; 6 and 7; 6 and 8; 6 and 5; 7 and 5; 7 and 8; and 5 and 8.*

For the given graph:
Edge connectivity:
To disconnect the graph, we need to remove all edges between the left retangel (vertices 1-4) and the right triangle (vertices 5-8).
There are 3 such edges connecting vertex 6 to vertices 2,3, and 4.
Therefore, $\kappa'(G) = 3$
Vertex connectivity:
Removing vertex 6 disconnects the graph into two components: the rectangle (vertices 1-4) and the triangle (vertices 5,7,8).
Therefore, $\kappa(G) = 1$.

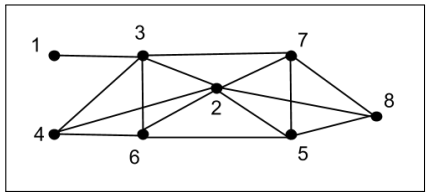
(ii) 

Figure 8: *An undirected graph has 8 vertices, 1 through 8. 4 vertices form a rectangular shape in the center. Starting from the top left vertex and moving clockwise, the vertices of the rectangular shape are, 3, 7, 5, and 6. Vertex 2 is at about the center of the rectangular shape. Vertex 8 is to the right of the rectangular shape. Vertex 1 and 4 are to the left of the rectangular shape, horizontally in-line with vertices 3 and 6, respectively. Undirected edges, line segments, are between the following vertices: 1 and 3; 3 and 7; 3 and 4; 3 and 6; 3 and 2; 4 and 2; 4 and 6; 6 and 2; 6 and 5; 2 and 5; 2 and 7; 2 and 8; 7 and 5; 7 and 8; and 5 and 8.*

For the graph in Figure 8:
Edge connectivity:
The minimum number of edges that need to be removed to disconnect the graph is 3.
For example, removing edges (1,3),(4,3), and (4,6) isolates vertex 1 from the rest of the graph.
Therefore, $\kappa'(G) = 3$.
Vertex connectivity:
The minimum number of vertices that need to be removed to disconnect the graph is 2. For example, removing vertices 3 and 6 disconnects vertex 1 and vertex 4 from the rest of the graph.
Therefore, $\kappa(G) = 2$.

Problem 4

For parts (a) and (b) below, find an Euler circuit in the graph or explain why the graph does not have an Euler circuit.

(a)

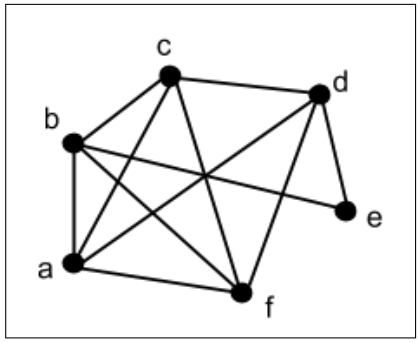


Figure 9: *An undirected graph has 6 vertices, a through f. 5 vertices are in the form of a regular pentagon, rotated 90 degrees clockwise. Hence, the top vertex becomes the rightmost vertex. From the bottom left vertex, moving clockwise, the vertices in the pentagon shape are labeled: a, b, c, e, and f. Vertex d is above vertex e, below and to the right of vertex c. Undirected edges, line segments, are between the following vertices: a and b; a and c; a and d; a and f; b and f; b and c; b and e; c and d; d and e; and d and f. Edges c f, a d, and b e intersect at the same point.*

A graph has a Euler circuit if and only if every vertex has even degree and the graph is connected.

For the graph in Figure 9,

I'll check the degree of each vertex:

-deg(a)=4
-deg(b)=4
-deg(c)=4
-deg(d)=3
-deg(e)=3
-deg(f)=4

Since vertices d and e have odd degrees, the graph does not have an Euler circuit.

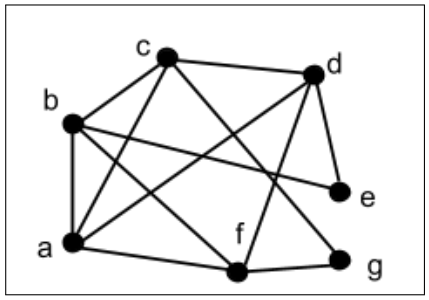
(b) 

Figure 10: *An undirected graph has 7 vertices, a through g. 5 vertices are in the form of a regular pentagon, rotated 90 degrees clockwise. Hence, the top vertex becomes the rightmost vertex. From the bottom left vertex, moving clockwise, the vertices in the pentagon shape are labeled: a, b, c, e, and f. Vertex d is above vertex e, below and to the right of vertex c. Vertex g is below vertex e, above and to the right of vertex f. Undirected edges, line segments, are between the following vertices: a and b; a and c; a and d; a and f; b and f; b and c; b and e; c and d; c and g; d and e; d and f; and f and g.*

For the graph in Figure 10,
I'll check the degree of each vertex:
-deg(a)=4
-deg(b)=4
-deg(c)=4
-deg(d)=4
-deg(e)=2
-deg(f)=4
-deg(g)=2
All vertices have even degree and the graph is connected.
Therefore, and Euler circuit exists.
One possible Euler circuit is:
a-b-c-a-d-c-g-f-a-f-d-e-b-f-b-a

(c) For each graph below, find an Euler trail in the graph or explain why the graph does not have an Euler trail.

(Hint: One way to find an Euler trail is to add an edge between two vertices with odd degree, find an Euler circuit in the resulting graph, and then delete the added edge from the circuit.)

(i)

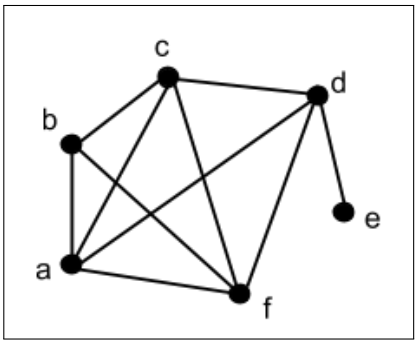


Figure 11: *An undirected graph has 6 vertices, a through f. 5 vertices are in the form of a regular pentagon, rotated 90 degrees clockwise. Hence, the top vertex becomes the rightmost vertex. From the bottom left vertex, moving clockwise, the vertices in the pentagon shape are labeled: a, b, c, e, and f. Vertex d is above vertex e, below and to the right of vertex c. Undirected edges, line segments, are between the following vertices: a and b; a and c; a and d; a and f; b and f; b and c; c and d; c and f; d and e; and d and f.*

A graph has an Euler trail if and only if it has exactly 0 or 2 vertices with odd degree, and the graph is connected.
For the graph in Figure 11,
I'll check the degree of each vertex:
-deg(a)=4
-deg(b)=3
-deg(c)=4
-deg(d)=4
-deg(e)=2
-deg(f)=5
Vertices with odd degree: b
(deg=3) and f (deg=5)
Since there are exactly 2 vertices with odd degree and the graph is connected, an Euler trail exists.
The Euler trail must start at one of the odd-degree vertices and end at the other.
One possible Euler trail is:
b-a-c-b-f-c-d-a-f-d-e-f

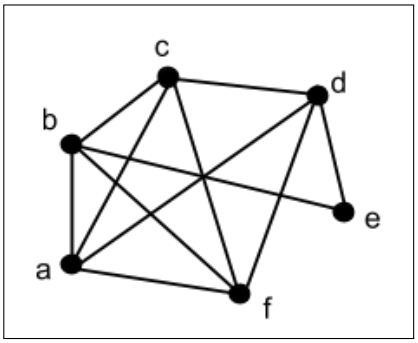
(ii) 

Figure 12: *An undirected graph has 6 vertices, a through f. 5 vertices are in the form of a regular pentagon, rotated 90 degrees clockwise. Hence, the top vertex becomes the rightmost vertex. From the bottom left vertex, moving clockwise, the vertices in the pentagon shape are labeled: a, b, c, e, and f. Vertex d is above vertex e, below and to the right of vertex c. Undirected edges, line segments, are between the following vertices: a and b; a and c; a and d; a and f; b and f; b and c; b and e; c and d; d and e; and d and f. Edges c f, a d, and b e intersect at the same point.*

For the graph in Figure 12,
I'll check the degree of each vertex:
-deg(a)=4
-deg(b)=4
-deg(c)=3
-deg(d)=4
-deg(e)=3
-deg(f)=4
Vertices with odd degree: c
(deg=3) and e (deg=3)
Since there are exactly 2 vertices with odd degree and the graph is connected, an
Euler trail exists.
One possible Euler trail is:
c-a-b-c-d-a-f-b-e-d-f-e

Problem 5

Consider the following tree for a prefix code:

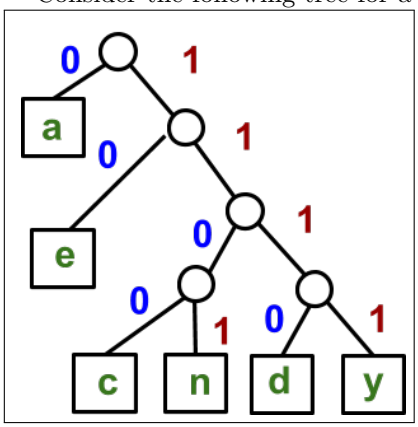


Figure 13: *A tree with 5 vertices. The top vertex branches into character, a, on the left, and a vertex on the right. The vertex in the second level branches into character, e, on the left, and a vertex on the right. The vertex in the third level branches into two vertices. The left vertex in the fourth level branches into character, c, on the left, and character, n, on the right. The right vertex in the fourth level branches into character, d, on the left, and character, y, on the right. The weight of each edge branching left from a vertex is 0. The weight of each edge branching right from a vertex is 1.*

(a) Use the tree to encode "day".

-d: Go right (1), right (1), right (1), left(0)-1110
-a: Go left (0)-0
-y: Go right (1), right (1), right (1), right (1)-1111
Therefore, the encoding of "day" is: 1110 0 1111

(b) Use the tree to encode "candy".

-c: Go right(1), right(1),left(0),left(0)-1100
-a: Go left (0)-0
-n: Go right (1),right(1),left(0),right(1-1101
-d: Go right (1), right(1),right(1),left(0)-1110
-y: Go right(1), right(1),right(1),right(1)-1111
Therefore, the encoding of "candy" is: 1100 0 1101 1110 1111

(c) Use the tree to decode “1110101101″.

-1110: Right, right, right, left -d
-10: Right, left-e
-1101: Right, right, left, right-n
Therefore, the decoded message is: ”den”

(d) Use the tree to decode “111001101110010″.

-1110: Right, right, right, left-d
-0: Left-a
-1101: Right, right, left, right-n
-1100: Right,right,left, left,-c
-1000: (incorrect code-doesn’t lead to a leaf)
There’s an error in the code.
The last part ”1000” doesn’t correspond to any character in the given prefix code. The decoded message is ”danc” followed by an error.

Problem 6

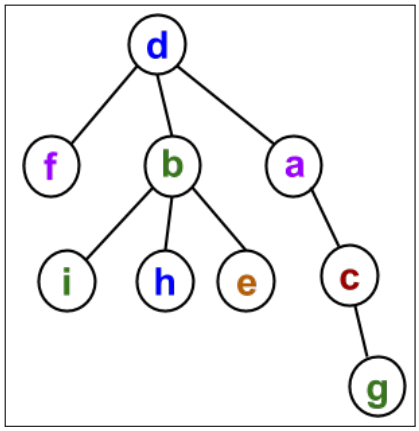


Figure 14: *A tree diagram has 9 vertices. The top vertex is d. Vertex d has three branches to vertices, f, b, and a. Vertex b branches to three vertices, i, h, and e. Vertex a branches to vertex c. Vertex c branches to vertex g.*

(a) Give the order in which the vertices of the tree are visited in a post-order traversal.

1. Start at the root d
2. Visit the left subtree f (it's a leaf): f
3. Visit the middle subtree b:
a. Visit i: i
b. Visit h: h
c. Visit e: e
d. Visit b: b
4. Visit the right subtree a:
a. Visit c:
i. Visit g: g
ii. Visit c: c
b. Visit a: a
5. Visit d: d
Therefore, the post-order traversal is:
f,i,h,e,b,g,c,a,d

(b) Give the order in which the vertices of the tree are visited in a pre-order traversal.

1. Visit the root d: d
2. Visit the left subtree f: f

3. Visit the middle subtree b:
a. Visit b: b
b. Visit i: i
c. Visit h: h
d. Visit e: e
4. Visit the right subtree a:
a. Visit a: a
b. Visit c: c
c. Visit g: g
Therefore, the pre-order traversal is: d,f,b,i,h,e,a,c,g

Problem 7

Consider the following tree. Assume that the neighbors of a vertex are considered in alphabetical order.

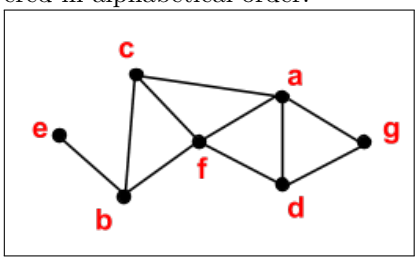


Figure 15: *A graph has 7 vertices, a through g, and 10 edges. Vertex e on the left end is horizontally inline with vertex g on the right end. Vertex b is below and to the right of vertex e. Vertex c is above vertex e and to the right of vertex b. Vertex f is between and to the right of vertices c and b. Vertex f is horizontally inline with vertices e and g. Vertex a is above and to the right of vertex f. Vertex d is below and to the right of vertex f. Vertex a is vertically inline with vertex d. Vertex g is between and to the right of vertices a and d. The edges between the vertices are as follows: e and b; b and c; c and f; c and a; a and d; b and f; f and a; f and d; a and g; and d and g.*

(a) Give the tree resulting from a traversal of the graph below starting at vertex a using BFS.

1. Visit a and mark it as visited
2. Enqueue all unvisited neighbors of a in alphabetical order: c, d, f, g
3. Dequeue c, mark it as visited
4. Enqueue all unvisited neighbors of c: b (f is already in the queue)
5. Dequeue d, mark it as visited
6. No new unvisited neighbors of d
7. Dequeue f, mark it as visited
8. No new unvisited neighbors of f
9. Dequeue g, mark it as visited
10. No new unvisited neightbors of g
11. Dequeue b, mark it as visited
12. Enqueue all unvisited neighbors of b: e
13. Dequeue e, mark it as visited

The resulting BFS tree has edges:
-(a,c)
-(a,d)
-(a,f)
-(a,g)
-(c,b)
-(b,e)

(b) Give the tree resulting from a traversal of the graph below starting at vertex a using DFS.

1. Visit a and mark it as visited
2. Visit the first unvistied neighbor of a in alphabetical order: c
3. Visit the first unvisited neighbor of c in alphabetical order: b
4. Visit the first unvisited neighbor of b in alphabetical order: e
5. e has no unvisited neighbors, backtrack to b
6. b has no more unvisited neighbors, backtrack to c
7. Visit the unvisited neighbor of c: f
8. Visit the first unvisited neighbor of f: d
9. d has no unvisited neighbors, backtrack to f
10. f has no more unvisited neighbors, backtrack to c
11. c has no more unvisited neighbors, backtrack to a
12. Visit the next unvisited neighbor of a: g
13. g has no unvisited neighbors, backtrack to a
14. a has no more unvisited neighbors, DFS complete

The resulting DFS tree has edges:
-(a,c)
-(c,b)
-(b,e)
-(c,f)
-(f,d)
-(a,g)

Problem 8

An undirected weighted graph G is given below:

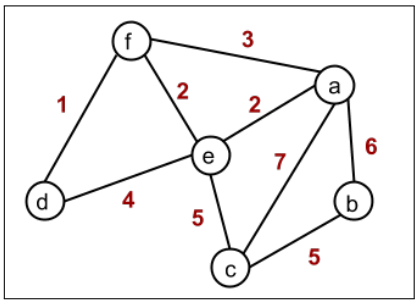


Figure 16: *An undirected weighted graph has 6 vertices, a through f, and 9 edges. Vertex d is on the left. Vertex f is above and to the right of vertex d. Vertex e is below and to the right of vertex f, but above vertex d. Vertex c is below and to the right of vertex e. Vertex a is above vertex e and to the right of vertex c. Vertex b is below and to the right of vertex a, but above vertex c. The edges between the vertices and their weight are as follows: d and f, 1; d and e, 4; f and e, 2; e and a, 2; f and a, 3; e and c, 5; c and a, 7; c and b, 5; and a and b, 6.*

(a) Use Prim's algorithm to compute the minimum spanning tree for the weighted graph. Start the algorithm at vertex a. Show the order in which the edges are added to the tree.

1. Start with vertex a
2. Consider all edges from a: (a,e) with weight 2, (a,f) with weight 3, (a,b) with weight 6, (a,c) with weight 7
3. Add the edge with minimum weight: (a,e) with weight 2
4. Consider all edges from a and e: (a,f) with weight 3, (e,c) with weight 5, (a,b) with weight 6, (a,c) with weight 7
5. Add the edge with minimum weight: (a,f) with weight 3
6. Consider all edges from a, e, and f: (e,c) with weight 5, (a,b) with weight 6, (a,c) with weight 7
7. Add the edge with minimum weight: (f,d) with weight 1
8. Consider all edges from a, e, f, and d: (e,c) with weight 5, (a,b) with weight 6, (a,c) with weight 7
9. Add the edge with minimum weight: (e,c) with weight 5
10. Consider all edges from a, e, f, d, and c: (c,b) with weight 5, (a,b) with weight 6
11. Add the edge with minimum weight: (c,b) with weight 5

The edges are added in this order: (a,e), (a,f), (f,d), (e,c), (c,b)

The minimum spanning trees has total weight: 2+3+1+5+5=16

(b) What is the minimum weight spanning tree for the weighted graph in the previous question subject to the condition that edge $\{d, e\}$ is in the spanning tree?

1. Include edge (d,e) with weight 4 in the MST

2. Apply Prim's algorithm to find the remaining edges, starting with vertices d and e

Remaining edges added in order:

-(e,a) with weight 2

-(a,f) with weight 3

-(e,c) with weight 5

-(c,b) with weight 5

The minimum spanning tree with edge d,e included has edges: (d,e), (e,a), (a,f), (e,c), (c,b) wiht total weight: 4+2+3+5+5=19

(c) How would you generalize this idea? Suppose you are given a graph G and a particular edge $\{u, v\}$ in the graph. How would you alter Prim's algorithm to find the minimum spanning tree subject to the condition that $\{u, v\}$ is in the tree?

1. Start by including the edge u,v in the MST

2. Contract the vertices u and v into a single vertex (call it uv)

3. Update the edges: For an edge that connected to either u or v, now connect it to the contract vertex uv (if both endopoints were u and v, remove the edge)

4. If there are multiple edges between uv and another vertex, keep on the one with minimum weight

5. Run Prim's algorithm on the modified graph

6. Expand the contracted vertex back into u and v, including the edgeu,v in the final MST

Alternatively a simpler approach:

1. Include edge u,v in the MST

2. Apply Prim's alogorithm on the original graph, but start with both vertices u and v already in the tree

3. Continue adding edges with minimum weight that connect vertices not yet in the tree to vertices already in the tree

4. This ensure that edge u,v is included while the rest of the MST is build optimally

This approach gurantees the minimum spanning tree subject to the constraint that u,v is included